

## SEMESTER S7

### ROBOTICS

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>OEMET722</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None            | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. This course helps the student to the basic idea of Robots. Students are introduced to the basic design considerations of robots.
2. Concepts like trajectory planning and obstacle avoidance and kinematics of robots are introduced.
3. Discussion on various mobile robots and robotic manipulators are also included as part of the course to get an overall idea on robotics

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Contact Hours</b> |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots.<br>Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers. | <b>9</b>             |
| <b>2</b>          | Direct Kinematics- Rotations-Fundamental and composite Rotations, Homogeneous coordinates, Translations and rotations, Composite homogeneous transformations, Screw transformations, Kinematic parameters, The Denavit-Hartenberg (D-H) representation, The arm equation, direct kinematics problems (up to 3DOF).<br>Inverse kinematics- general properties of solutions and problems (up to                                                                                                           | <b>10</b>            |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | 3DOF).                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
| <b>3</b> | Manipulator Dynamics: Lagrange's formulation – Kinetic Energy expression, velocity Jacobian and Potential Energy expression, Generalised force, Euler-Lagrange equation, Dynamic model of planar and spatial serial robots up to 2 DOF, modelling including motor and gearbox.                                                                                                                                                    | <b>8</b> |
| <b>4</b> | Trajectory Planning. Joint space trajectory planning- cubic polynomial, linear trajectory with parabolic blends, trajectory planning with via points; Cartesian space planning, point-to-point vs. continuous path planning. Obstacle avoidance methods- Artificial Potential field, A* algorithms.<br>Robot Control: The control problem, Single axis PID control-its disadvantages, PD gravity control, computed torque control | <b>9</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| <b>Part A</b>                                                                                                                                                                                 | <b>Part B</b>                                                                                                                                                                                                                                                                                                | <b>Total</b> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>2 Questions from each module.</li> <li>Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>Each question carries 9 marks.</li> <li>Two questions will be given from each module, out of which 1 question should be answered.</li> <li>Each question can have a maximum of 3 sub divisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                                             | Bloom's Knowledge Level (KL) |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Familiarise with anatomy, specifications and types of Robots                                                                | <b>K2</b>                    |
| <b>CO2</b>     | Obtain forward and inverse kinematic models of robotic manipulators                                                         | <b>K3</b>                    |
| <b>CO3</b>     | Plan trajectories in joint space & Cartesian space and avoid obstacles while robots are in motion                           | <b>K4, K5</b>                |
| <b>CO4</b>     | Develop a dynamic model and design the controller for robotic manipulators                                                  | <b>K4, K6</b>                |
| <b>CO5</b>     | Choose the appropriate Robotic configuration and list the technical specifications for robots used in different application | <b>K4</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 2   | 1   | -   | -   | -   | -   | -   | -   | -   | -    | -    | 3    |
| <b>CO2</b> | 2   | 1   | -   | -   | -   | -   | -   | -   | -   | -    | -    | 3    |
| <b>CO3</b> | 2   | 1   | -   | -   | -   | -   | -   | -   | -   | -    | -    | 3    |
| <b>CO4</b> | 3   | 2   | 2   | -   | -   | -   | -   | -   | -   | -    | -    | 3    |
| <b>CO5</b> | 3   | 2   | 2   | -   | -   | -   | -   | -   | -   | -    | -    | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                                  |                             |                              |                         |
|-------------------|--------------------------------------------------|-----------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                         | <b>Name of the Author/s</b> | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                 | Fundamentals of Robotics – Analysis and Control  | Robert. J. Schilling        | Prentice Hall of India       | 1996                    |
| 2                 | Introduction to Robotics (Mechanics and Control) | John. J. Craig              | Pearson Education Asia       | 2002                    |
| 3                 | Introduction to Robotics                         | S K Saha,                   | McGraw Hill Education        |                         |
| 4                 | Robotics and Control                             | R K Mittal                  | Tata McGraw Hill, New Delhi  | 2003                    |
| 5                 | Robotics-Fundamental concepts and analysis       | AshitavaGhosal              | Oxford University Press      |                         |
| 6                 | Robotics Technology and Flexible Automation,     | S. R. Deb                   |                              | Second Edition,         |
| 7                 | Introduction to Autonomous Mobile Robots         | Siegwart, Roland,           | Cambridge, Mass.: MIT Press, | Second Edition,         |

| <b>Reference Books</b> |                                                  |                                                                    |                              |                         |
|------------------------|--------------------------------------------------|--------------------------------------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                         | <b>Name of the Author/s</b>                                        | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                      | Handbook of Robotics                             | Siciliano, Khatib                                                  | Springer                     |                         |
| 2                      | Introduction to Robotics – Mechanics and Control | John J. Craig                                                      |                              |                         |
| 3                      | Modern Robotics Mechanics, Planning and Control  | . Kevin M. Lynch, Frank C. Park,                                   |                              |                         |
| 4                      | Robotics Modelling, Planning and Control         | Bruno Siciliano, Lorenzo Sciavicco, Luigi Villani, Giuseppe Oriolo | Springer                     |                         |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                                                     |
| 1                              | <a href="https://archive.nptel.ac.in/courses/112/105/112105249/">https://archive.nptel.ac.in/courses/112/105/112105249/</a> |
| 2                              | <a href="https://archive.nptel.ac.in/courses/112/105/112105249/">https://archive.nptel.ac.in/courses/112/105/112105249/</a> |
| 3                              | <a href="https://archive.nptel.ac.in/courses/112/105/112105249/">https://archive.nptel.ac.in/courses/112/105/112105249/</a> |
| 4                              | <a href="https://archive.nptel.ac.in/courses/112/105/112105249/">https://archive.nptel.ac.in/courses/112/105/112105249/</a> |